AUSTERE MEDICAL & SURVIVAL FIELD MANUAL: CHEAT-SHEET

Tactical Quick Reference • Prolonged Field Care (PFC) • Disaster Medicine Protocol

1. AIRWAY MANAGEMENT: MSMAID CHECKLIST & SURGICAL CRICOTHYROIDOTOMY

Domain	Minimum	Better	Best
Machine	Ventilation / BVM + PEEP valve	Portable ventilator	Full feature vent + O2 concentrator
Suction	Syringe + NPA / bulb suction	Manual foot suction	Powered commercial suction unit
Monitor	Pulse Oximetry (SpO2)	EtCO2 colorimetric / Capnometry	Continuous Waveform Capnography
Airway	NPA / Cricothyroidotomy Kit	Supraglottic Airway (i-gel/King)	Videolaryngoscopy / ETT + cuffed tube
IV / IO Access	18G IV or IO catheter	Dual IV/IO access	Central line / Rapid infuser
Drugs	Procedural Ketamine / Lidocaine	Analgesia + Sedation protocol	Continuous IV infusion + paralytics

SURGICAL AIRWAY INDICATIONS: Massive facial/neck trauma, expanding hematoma, inhalation burns/edema, anaphylaxis, 'Cannot Ventilate/Cannot Intubate'.
CONTRAINDICATIONS: Children < 12 years old (use BVM/SGA, never surgical cric); Simple unconsciousness; Traumatic cardiac arrest.

2. TOURNIQUET CONVERSION & MASS CASUALTY TRIAGE (START / JUMPSTART)

Category	Protocol & Clinical Rules
< 2 Hours	**TQ Conversion Window:** Attempt if no shock, wound monitorable, no amputation. Place 'Plus-1' TQ above original, pack hemostatic gauze 3 min, apply pressure dressing, slowly release original TQ.
2 - 6 Hours	**High-Risk Conversion:** Attempt conversion only with experienced provider present and full resuscitation capabilities. Monitor for re-bleeding.
> 6 Hours	**DEAD LIMB HARD LIMIT:** DO NOT remove or loosen TQ in field! Releasing floods systemic circulation with K+, lactic acid, & myoglobin (fatal reperfusion injury).
START Triage	**RPM Filter: Green:** Walking wounded. | **Black:** Unresponsive & no respirations after 1 airway opening. | **Red:** Respirations >30 or <10 | Cap refill >2s / no radial pulse | Cannot follow commands. | **Yellow:** Passed RPM checks.
JumpSTART (Ped)	Pediatric modification: If child is apneic but has a pulse, give **5 rescue breaths** before tagging Black.

3. EXPIRED PHARMACEUTICALS, SLEP STANDARDS & DOXYCYCLINE

Topic	Survival Field Guidelines
SLEP 88% Rule	FDA Shelf Life Extension Program showed **88% of solid meds** retain full potency & safety ≥1 year past expiration (average extension: 66 months / 5.5 years).
Doxycycline Myth	Modern doxycycline **DOES NOT** degrade into nephrotoxic compounds (1960s Fanconi myth). Expired doxycycline slowly loses antimicrobial potency; it does not poison kidneys.
Discard Criteria	Discard immediately if liquid/injectable is cloudy, discolored, or contains sediment; or if solid pills show crumbling or severe chemical odor.

4. IMPROVISED HYDRATION PROTOCOLS & PROCTOCLYSIS

Protocol	Clinical Mechanism & Mixing Ratios
SGLT1 Pump	Sodium-Glucose Cotransporter 1 actively binds 2 Na+ and 1 Glucose, driving active water absorption across gut mucosa during diarrheal illness.
WHO ORS Recipe	**1.0L water** + **1/2 tsp salt (2.6g NaCl)** + **6 tsp sugar (13.5-20g sucrose)** + Potassium (1/2 mashed banana, 1/2 cup OJ, or 1/4 tsp KCl).
PFC Recipe	**1.0L water** + 8 tsp sugar + 1/2 tsp salt + 1/2 tsp baking soda (sodium bicarbonate).
Sports Drink Hazard	Unmodified Gatorade/soda is hyperosmolar (>300-500 mOsm/L, low Na+). Reverses intestinal fluid gradient, causing severe osmotic diarrhea. Add 1/4 tsp salt & 1/4 tsp baking soda per quart.
Slow-Sip Protocol	Give 1 tbsp (15 mL) q5min. If held down 20 min, scale to 2 tbsp (30 mL) q5min. Target UOP: >180 mL per 6 hours (0.5-1.5 mL/kg/hr).
Proctoclysis	Murphy Drip: Insert catheter 10-15 cm rectally, elevate ORS 30-50 cm above patient. Gravity drip at 200-500 mL/hr for systemic absorption.

5. IMPROVISED WOUND CARE, DAKIN'S SOLUTION & BIO-HACKS

Intervention	Technique, Dilution & Bio-Hack Protocols
Golden Rule	**NEVER STITCH A DIRTY WOUND!** Perform Delayed Primary Closure (DPC) at Day 3-5 (or Day 4-7) only after clean granulation forms.
Irrigation	1-3L (small), 4-8L (medium), 9+L (large/gross). Use 19G catheter + 35 mL syringe (8 psi, ideal 6-15 psi range). Boil water 3 min if unpotable.
Dakin's Solution	**Half-Strength Stock:** 1L water + 5 mL household bleach (5.25% NaOCl) + 1.5 mL baking soda. **Wound Flushing:** Dilute stock **1:10 with water** (0.025% final strength).
Sugar / Honey	Pack wound cavity with raw honey or granulated sugar. High osmotic gradient starves bacteria of moisture and pulls out infection.
Sphagnum Moss	Clean & boil wetlands moss. *Sphagnan* cell wall polymer binds Na+/K+ and releases H+, acidifying wound bed to **pH 3.5-4.5** to kill *Staph* & *Pseudomonas*.
Maggot Therapy	Green bottle fly larvae (*Lucilia sericata*) at 5-10 larvae/cm^2 for 48-72 hrs under mesh dressing to digest dead tissue and spare living flesh.
Sterilization	Submerge tools in 5% formaldehyde solution (1-24h), undiluted 5.25% bleach (5 min), or boil in water (1 min minimum, last resort).

6. AUSTERE PAIN MANAGEMENT & TRIPLE-OPTION ANALGESIA

Tier	Analgesic Protocol & Dosing
Goal	Tolerable pain, intact mentation, preserved airway drive. Start low, go slow!
Option 1 (Mild)	**CWMP Pill Pack:** Acetaminophen 1,000 mg + Meloxicam 15 mg PO daily. No mental clouding or bleeding risk.
Option 2 (Mod/No Shock)	**OTFC Fentanyl 800 mcg lozenge** transmucosal (tape to finger, between cheek & gum). Or Fentanyl IV 25-50 mcg / IN 50-100 mcg.
Option 3 (Shock / TBI)	**Ketamine 20-30 mg slow IV/IO over 1 min** (or 50-100 mg IM/IN) q20-30min. Maintains BP & airway reflexes.
Procedural Sedation	**Ketamine 1-2 mg/kg slow IV/IO** (or 300 mg IM). **DISARM CASUALTY FIRST!** Administer Midazolam 0.5-2 mg IV for emergence delirium.
Naloxone Reversal	Dilute 0.4 mg Naloxone in 9 mL NS (40 mcg/mL). Push **40 mcg slowly** to restore breathing without stripping all analgesia.

7. TACTICAL FIELD SPLINTING & COMPARTMENT SYNDROME

Domain	Clinical Guidelines & Surgical Warning Signs
Splinting Rules	TFC only (never in CUF). Immobilize **joint above and joint below**. Perform CMS check before and after. Realign bone ONLY if pulseless (1 gentle attempt). Pad bony prominences.
Compartment 6 Ps	**1. Pain out of proportion** (earliest/key sign!) | **2. Paresthesia** | **3. Pallor** | **4. Paralysis** | **5. Poikilothermia** | **6. Pulselessness** (late sign).
Interventions	Cool limb evaporatively (DO NOT pack in ice). Keep limb at heart level (DO NOT elevate). Escharotomy for circumferential burns; Fasciotomy for deep muscle pressure.
Dead Limb Rule	Cool, insensate, pulseless extremity = dead muscle. Place **2 side-by-side TQs proximal** to isolate toxins and **DO NOT remove**!

8. CRUSH SYNDROME & 4-STEP HYPERKALEMIA EMERGENCY PROTOCOL

Step / Phase	Action, Drug Dosing & Critical Safety Warnings
Reperfusion Injury	Traumatic rhabdomyolysis flushes K+, myoglobin, & lactic acid into blood upon unpinning, causing fatal cardiac arrest or renal failure.
Pre-Extrication Fluids	Start IV/IO **BEFORE lifting debris**! Give **2L IV bolus**, run at 1-1.5 L/hr, target UOP 100-200 mL/hr (300 mL/hr first 24h).
Tourniquet Isolation	If trapped >2 hrs without fluids/meds available, place **2 TQs above crush site BEFORE extrication** to block toxic flush.
Step 1: Shield Heart	**10% Calcium Gluconate 10 mL IV over 2-3 min** (or Calcium Chloride 1g over 10 min) to stabilize cardiac membrane.
Step 2: Shift K+ Cells	**10 units Regular Insulin IV + 50 mL D50 (25g glucose)** + **Albuterol 10 mg neb** (or 15-20 puffs MDI) to push K+ intracellularly.
Step 3: Alkalinize / Protect	**Sodium Bicarbonate 50 mEq in 1L fluid** (max 200 mEq/24h). **WARNING: NEVER MIX CALCIUM & BICARBONATE IN SAME LINE!**
Step 4: Excrete K+	**Kayexalate 15-30g PO/PR** to bind and physically eliminate potassium from the gastrointestinal tract.